

Apex Planet Data Analytics Internship - Task 4

Data Storytelling & Statistical Validation

Product Performance & Sales Drivers

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Business Objective

The objective of this analysis is to understand the major factors influencing sales performance across products, cities, and time.

The analysis combines:

- Product performance analysis
- Transaction volume analysis
- Sales value analysis
- City-level performance
- Monthly sales trends
- Statistical validation of a key business hypothesis

Key Business Questions

1. Which products generate the highest sales?
2. Is sales performance influenced more by transaction volume or transaction value?
3. Which products have the highest units sold?
4. Which products have the highest average transaction value?
5. How does product performance vary across cities?
6. Does the observed difference in average transaction value between selected products show statistical significance?

Dataset Overview

The analysis uses a cleaned retail sales transaction dataset.

Dataset Details

- **Transactions:** 1,000
- **Original Columns:** 12
- **Final Analysis Columns:** 16
- **Products:** 6
- **Categories:** 5
- **Cities:** 9 including Unknown
- **Date Range:** January 2025 – January 2026
- **2025 Records:** 995
- **January 2026 Records:** 5

Key Variables

- **Order Date**
- **Customer ID**
- **Age**
- **Gender**
- **City**
- **Product**
- **Category**
- **Quantity**
- **Unit Price**
- **Total Sales**

Tools Used

- **Python**
- **Pandas**
- **NumPy**
- **Matplotlib**
- **Seaborn**
- **Power BI**

From Data to Business Insight

Approach: The analysis combines descriptive analytics, visualization, and statistical validation to move from raw transaction data to business insights.



Overall Sales Performance

Key Metrics

Total Sales: ₹139.40M

Transactions: 1,000

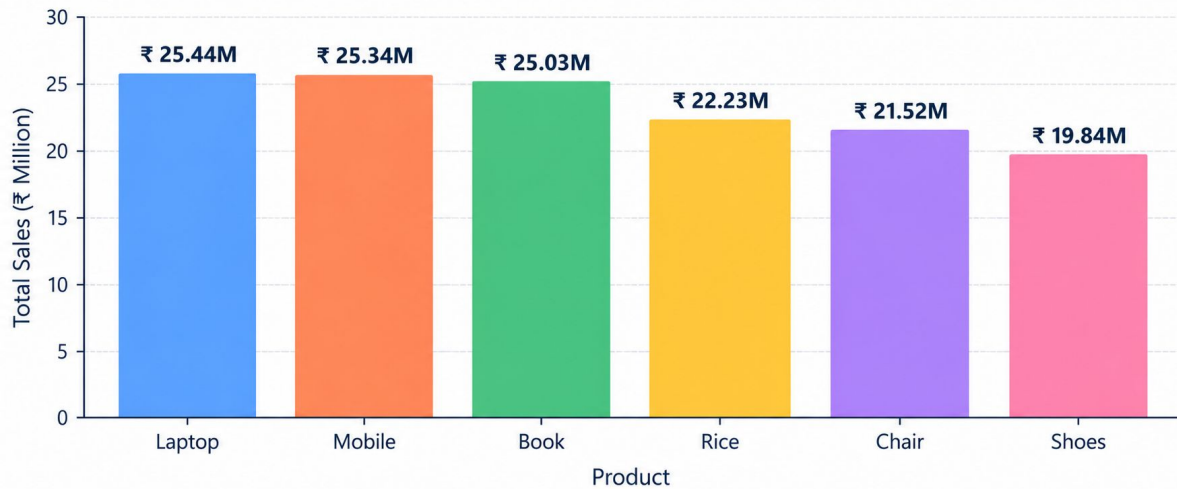
Average Transaction Value: ₹139,399

Median Transaction Value: ₹108,594

Key Observation

The mean transaction value is higher than the median, indicating that relatively high-value transactions increase the overall average.

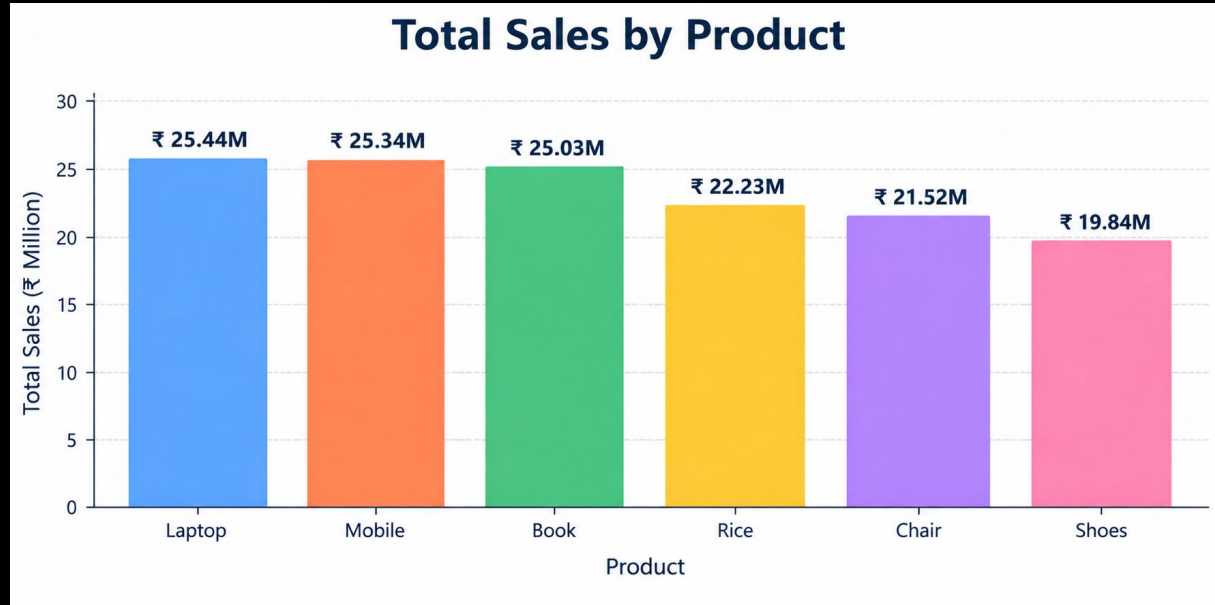
Total Sales by Product



Product Sales Performance

Key Insights

- Laptop recorded the highest total sales.
- Mobile was very close to Laptop in total sales.
- Book was the third-highest product by total sales.
- Shoes recorded the lowest total sales among the six products.
- Laptop, Mobile, and Book together contributed approximately **54.4%** of overall sales.



Sales Drivers: Volume vs Transaction Value

Product sales are influenced by both:

Transaction Volume

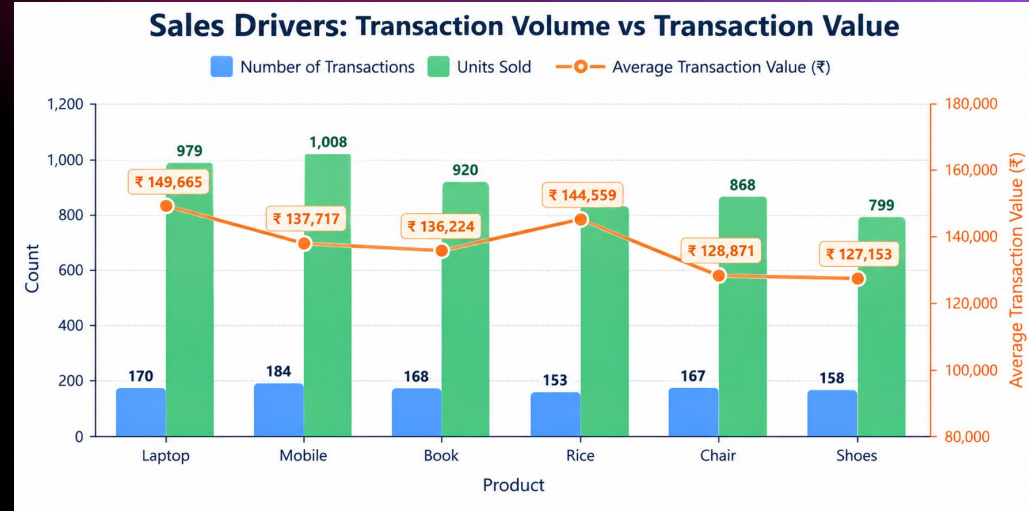
- Number of transactions
- Units sold

Transaction Value

- Average transaction value
- Unit price

Key Findings

- Mobile recorded the highest transaction count: **184**
- Rice recorded the lowest transaction count: **153**
- Mobile sold the highest number of units: **1,008**
- Shoes sold the lowest number of units: **799**
- Laptop had the highest average transaction value: approximately **₹149,665**
- Shoes had the lowest average transaction value: approximately **₹127,153**



Business Insight

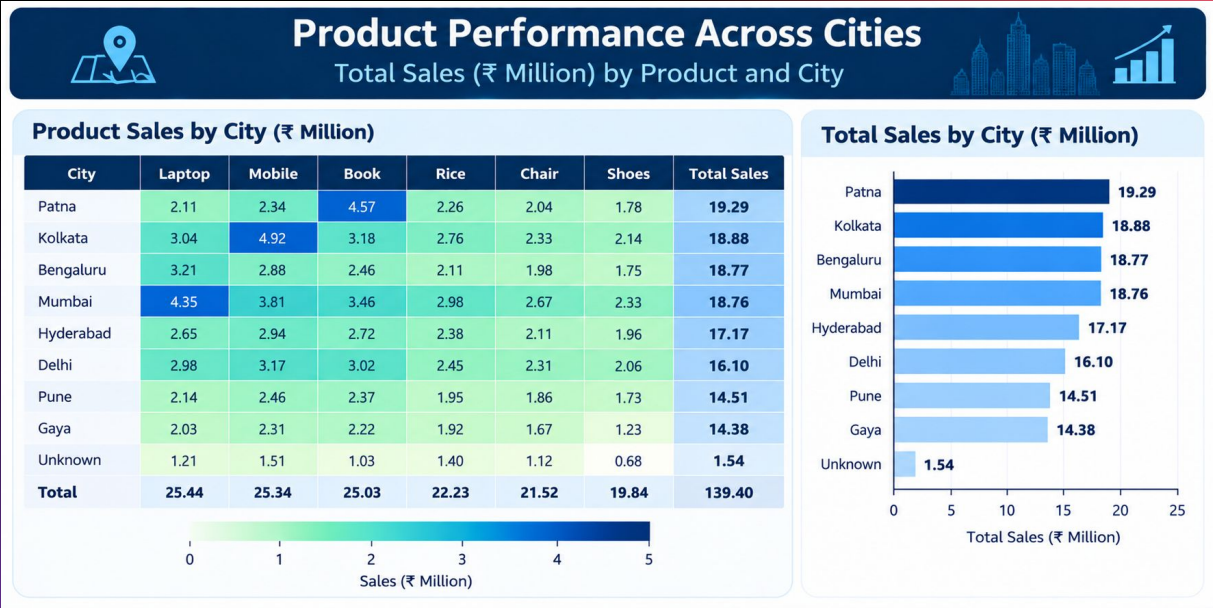
Higher transaction volume does not always result in the highest total sales. Product sales reflect a combination of transaction frequency, quantity, and transaction value.

Product Performance Across Cities

The analysis shows that product performance varies across locations.

Key Findings

- Mobile recorded its highest city-level sales in **Kolkata**, at approximately **₹4.92M**.
- Laptop showed strong sales in **Mumbai**, at approximately **₹4.35M**.
- Laptop also showed strong performance in **Bengaluru**, at approximately **₹3.21M**.
- Book recorded strong sales in **Patna**, at approximately **₹4.57M**.



Business Insight

Product performance is not uniform across cities. Location-level differences can help identify markets where individual products perform particularly well.

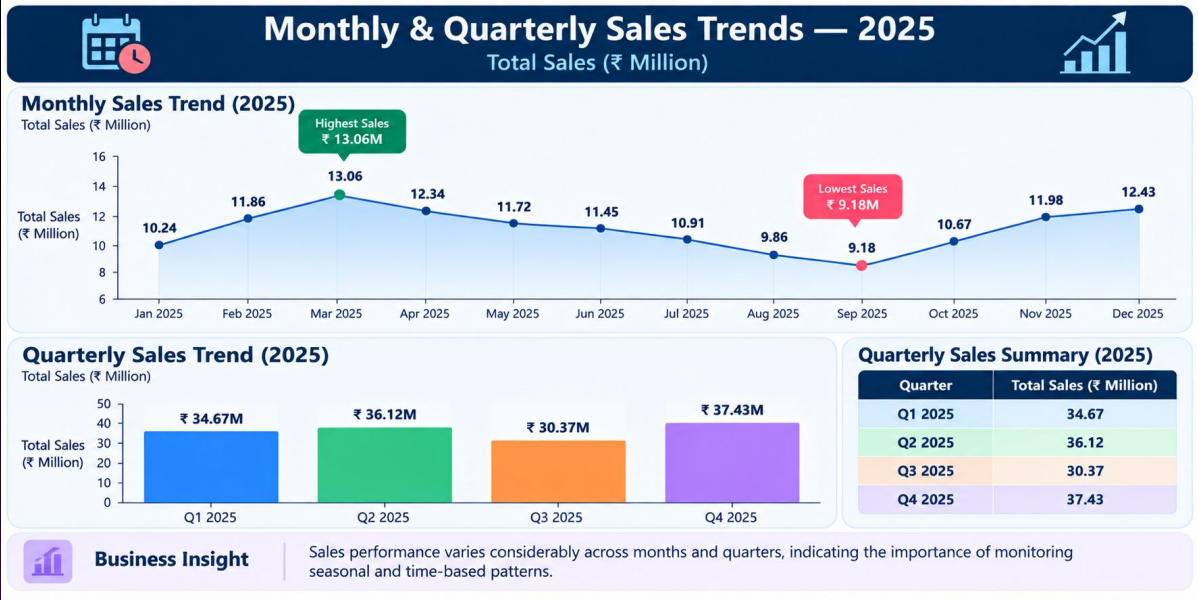
Monthly & Quarterly Sales Trends

Key Findings

- **March 2025** recorded the highest monthly sales at approximately **₹13.06M**.
- **September 2025** recorded the lowest monthly sales at approximately **₹9.18M**.
- **Q4 2025** recorded the highest quarterly sales at approximately **₹37.43M**.
- **Q3 2025** recorded the lowest quarterly sales at approximately **₹30.37M**.

Business Insight

Sales performance varies considerably across months and quarters, indicating the importance of monitoring seasonal and time-based patterns.



Business Hypothesis

Research Question

Is the average transaction value significantly different between Laptop and Mobile transactions?

Null Hypothesis (H_0) - There is no statistically significant difference in the average transaction value between Laptop and Mobile transactions.

Alternative Hypothesis (H_1) - There is a statistically significant difference in the average transaction value between Laptop and Mobile transactions.

Significance Level: $\alpha = 0.05$

Statistical Test - Welch's Independent Two-Sample t-test

Why this test?

The test compares the mean transaction values of two independent product groups while not requiring equal population variances.

Hypothesis Testing Results

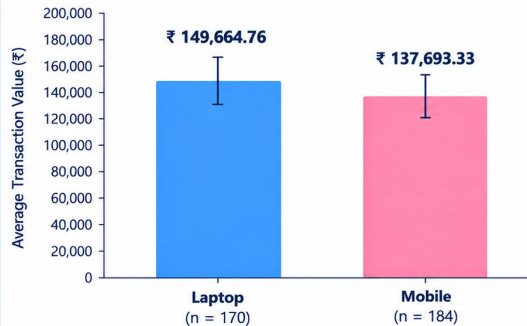
	Metric	Result
Decision Rule	Laptop Mean Transaction Value	₹149,664.76
	Mobile Mean Transaction Value	₹137,693.33
	Difference in Means	₹11,971.42
Test Result	t-statistic	0.954
	p-value	0.341
	95% Confidence Interval	₹-12,717.22 to ₹36,660.07
Business Interpretation		

The test does not provide sufficient statistical evidence that the average transaction value differs significantly between Laptop and Mobile transactions.

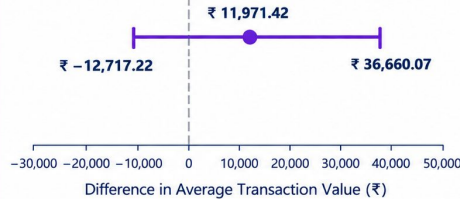
Statistical Interpretation

Average Transaction Value by Product

Average Transaction Value by Product



95% Confidence Interval for Difference in Means (Laptop – Mobile)



Business Interpretation

Although Laptop transactions have a higher observed average transaction value than Mobile transactions, the statistical test does not provide sufficient evidence that this difference is statistically significant.

Interpretation of the Statistical Test

The p-value is **0.341**, which is above the **0.05 significance level**. Therefore, the null hypothesis is **not rejected**.

The analysis does not provide sufficient statistical evidence of a difference in average transaction value between **Laptop and Mobile transactions**.

Confidence Interval

The **95% confidence interval** for the difference in average transaction value is:

₹ -12,717.22 to ₹ 36,660.07

The confidence interval provides the estimated range for the difference in average transaction values between the two product groups.

Key Business Insights

Laptop generated the highest total sales, while Mobile generated the highest transaction volume and units sold.

Transaction volume alone does not fully describe product sales performance; transaction value also contributes to differences in total sales.

Statistical testing adds rigor to descriptive findings
by determining whether an observed difference
between product groups is statistically supported.

Product performance varies across cities, with different products showing stronger sales in different locations.

Sales vary across time, with March recording the highest monthly sales and Q4 recording the highest quarterly sales.

Business Action Points

Based on the analysis, businesses can:

Monitor both **transaction volume** and **transaction value** when evaluating product performance.

Use monthly and quarterly trends to support sales planning.

Use statistical testing to validate important differences before making data-driven decisions.

Identify cities where specific products demonstrate stronger sales performance.

Investigate products with high transaction volume but comparatively different transaction values.

Continue monitoring product and location performance through interactive dashboards.

Conclusion

The analysis combined exploratory data analysis, product-level deep-dive analysis, interactive dashboarding, and statistical validation to understand retail sales performance.

The findings show that product performance is influenced by multiple factors, including transaction volume, quantity sold, transaction value, location, and time.

The statistical hypothesis test provides an additional level of evidence for evaluating differences between product groups.

Overall Outcome

The analysis transforms transaction data into actionable business insights and demonstrates how descriptive and statistical analysis can support data-driven decision-making.

THANK YOU